

Lily Myers

Objective

Aspiring astrophysicist and geophysicist utilizing machine learning and payload engineering to advance space science. Focused on leveraging rigorous undergraduate research and international collaboration to uncover cosmic origins while expanding inclusive STEM mentorship.

Contact Information

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 LinkedIn

Educational History

University of California, Berkeley

B.A. in Astrophysics and Geophysics, Minor in French | 2025 - 2029

GPA: **3.88/4.00**

- Member of CalRotaract, STAC (Space Technologies at Cal), and SWPS (Society of Women in Physical Sciences)

Eastlake High School

High School Diploma | 2021 - 2025

GPA: **3.96/4.00**

- Inducted Member of National Honor Society, Physics Honor Society, French Honor Society and Key Club
- Successfully completed **13 dual-enrollment college courses** at local community colleges

Employment & Internship History

SEES Zero-G Researcher

NASA & UT Austin Center for Space Research | Sept. 2024 - June 2025

- Selected as **1 of 4 interns** (from **80+ applicants**) for parabolic zero-g flight experiment
- Designed, constructed, and flew a **fluid dynamics experiment addressing fuel slosh in zero-gravity** under mentorship of experts
- Spoke at conferences and on podcasts to mentor and encourage high school students in STEM

SEES Onsite Intern

NASA & UT Austin Center for Space Research | July 2024

- Investigated asteroid composition to better understand asteroid formation and evolution
- Presented this research to NASA and at **American Geophysical Union (AGU) annual conference**

Extracurriculars

Participant

CosmoVerse Data Challenge | July 2026 - Present

- Calculating host galaxy distances using TRGB (Tip of the Red Giant Branch) from Type Ia supernovae to constrain H_0
- Planning on **scaling analysis beyond provided Jupyter notebooks** by designing a workflow pipeline using Python API queries, SQLite databases, and a Bash wrapper script

Chair of External Partnerships Committee + Financial Director

STAC (Space Technologies at Cal) | Mar. 2026 - Present

- Directing the committee alongside External Vice President in **cultivating industry partnerships**, securing launch opportunities and project funding
- Expanding professional resources to members through guest speakers, workshops, and database of internships/jobs

TIME II Mechanical Engineering Team Lead (Previously Team Member)

STAC (Space Technologies at Cal) | Aug. 2025 - Present

- Investigating microgravity's effects on growing E. coli by engineering a payload that tests gene expression in response to antibiotics
- **Directing team of about 10 engineers** to design, manufacture, and test our payload, iteratively refining it until ready for a Spring 2027 launch

Research Mentee

ULAB (Undergraduate Labs at Berkeley) | Aug. 2025 - May 2026

- Applied **machine learning techniques using PyTorch** to develop predictive models for gravitational microlensing curves, enhancing the accuracy of event forecasts and **optimizing data-processing efficiency** for large-scale surveys

Awards & Honors

Van Nattan-Hansen-Anderson Scholarship Recipient, Denver Astronomical Society
| Sept. 2025